



The Role of Sartorius' Lab Water System in Reduction of the Microplastic Content in Ultrapure Water

Ensuring Particle-Free Ultrapure Water Through Advanced Filtration

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Microplastics in water and the environment is one of the hot topics under the general discussion of environmental professionals. These microplastic particles are present everywhere, also in tap water. Microplastics can already be found inside human bodies and are suspected of causing long-term health problems. Now, the authorities and other professionals are working on new regulations to reduce the amount of microplastic load in the future.

This note describe show well the Sartorius Arium® Pro VF series can remove microplastic particles from normal tap water, deionized water or other suitable water sources using two different test setups and prevent the particles from causing any challenges in laboratory work.

Keywords or phrases:

Microplastic, μ -Raman, ASTM Type I water, Laboratory Water, Arium® Pro VF

Microplastics are tiny fragments (less than 5,000 to 1 μm in diameter)¹ of plastics that end up in nature by degradation on bigger plastic products or from sources like cosmetics, clothing, car tires, and industrial processes (Figure 1). There is rising concern about microplastic pollution in lakes, oceans, and drinking water, which in turn increases the demand for regulation and laboratory testing of microplastics in different kinds of products, environmental samples and other sample types since these particles can

eventually cause for example health hazards.²

Microplastics are found everywhere including tap water and other clean water sources, but also inside human bodies. The long-term health effect of microplastics is still unknown, but preliminary evidence suggests that they might affect stress levels and immune responses. For these reasons, it is really important to implement all possible solutions to hinder microplastic count in different locations, such as in laboratory water systems.^{3,4}

Many different types of water purification systems have been introduced to the laboratories, but microplastics are a topic that has not been focused on in the past. In general, microplastics levels in different water sources can be still unknown in many areas. Laboratories around the world should be able to be sure that, for example, all possible contamination sources are taken into account when using water in the laboratory.

Currently, new regulations and different standards are under development to get more tools to control the amount of microplastics in the environment. The latest of these is the European Chemical Agency's proposal to ban intentionally added microplastics in different products. The first restrictions of this regulation entered into force in the fall of 2023.⁵

Ultrapure Water Purification System Overview

Application-Oriented and Flexible to Meet the Highest Demands

The Arium® Pro VF series stands as a trusted source of ultrapure water, and it distinguishes itself by offering a flexible and modular system. In comparison to conventional devices, these systems provide remarkable added value.

One standout feature is that all systems within this series not only meet but exceed the stringent ASTM Type 1 water quality standards, ensuring the delivery of the highest quality results in their category. These systems are capable of dispensing up to 2 liters of consistently high-quality ultrapure water per minute, with a conductivity of 0.055 $\mu\text{S x cm}^{-1}$ (equivalent to 18.2 $\text{M}\Omega \times \text{cm}$). Moreover, when equipped with the Arium® Sterile Plus final filter,

they deliver ultrapure water that is virtually free from microorganisms.

The Arium® Pro series (Figure 2) embodies the following characteristics:

- ▶ **Modular.** Each system can be specifically selected to suit your application.
- ▶ **Flexible.** They seamlessly integrate into a variety of laboratory environments.
- ▶ **User-Friendly.** These systems feature a touch-activated display with an intuitive menu for easy operation.
- ▶ **Efficient.** With the favorites function, you gain direct access to recurring volumes, streamlining workflows, and saving valuable time in your laboratory.

Materials

Water samples were filtered using silicon filters with 1 µm pore size (Smart Membranes GmbH, Germany). 10, 25 and 70 µm NIST Traceable Size Standard polystyrene were used for spiking experiments. Particle analyses using Raman microscopy were performed with DXR3xi Raman Imaging Microscope. All spectra were collected using a 532 nm laser, filter, grating set, with the following conditions: 10 mW laser power, 0.05 s exposure time, and the number of spectra was 50. An Olympus magnification apochromatic objective with a 20x magnification was used and brightfield or darkfield illumination mode were used interchangeably for achieving optimal contrast in particle detection.

Method

This study was done with two main objectives in mind: the first objective was to investigate whether the Arium® device itself leached particles into the final ASTM type I water it produced (Figure 2), and the second objective was to see if the system retained the particles introduced to the system from the feed water (Figure 3).

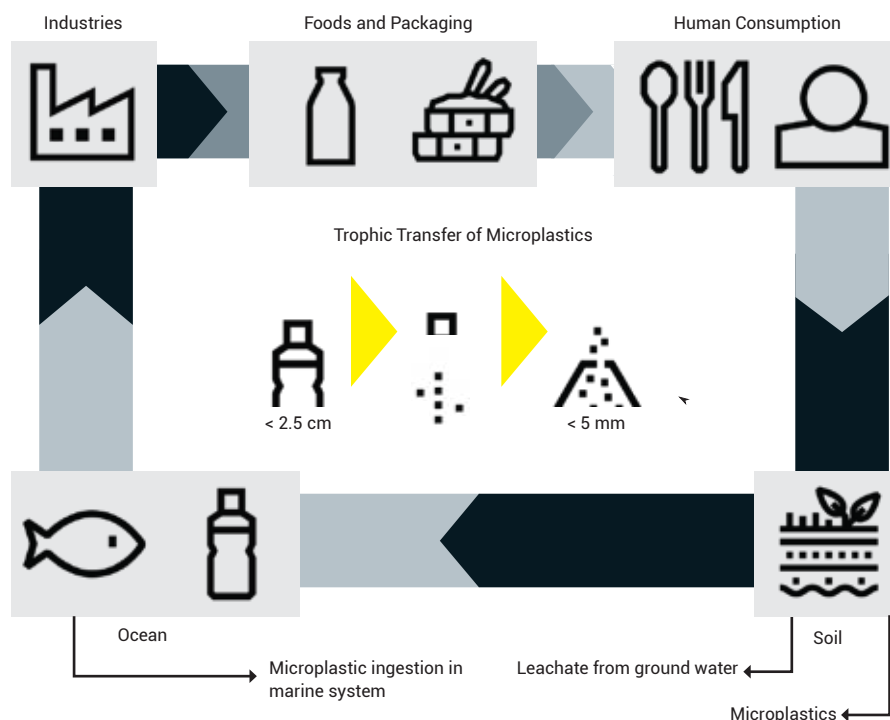


Figure 1: Major sources of microplastics and its transport into food systems.²

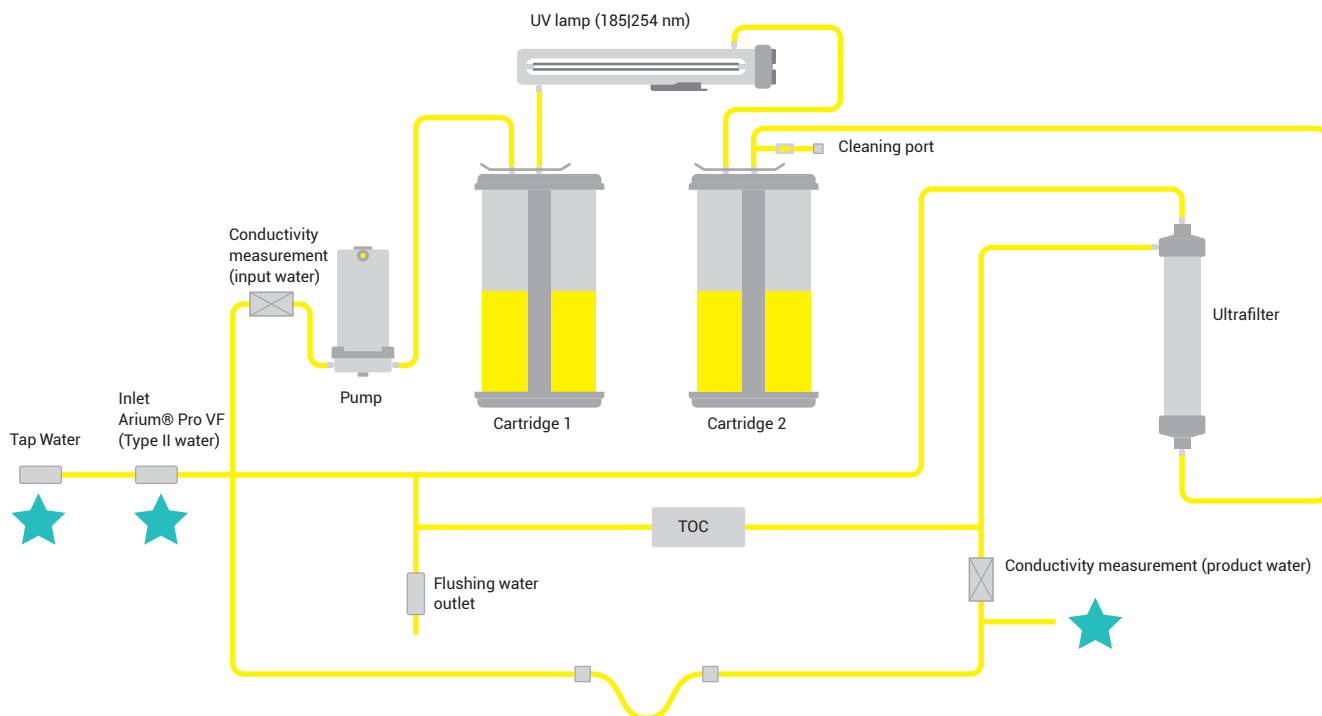


Figure 2: Flowchart of Arium® Pro VF for the second objective. Teal color stars mark the point of sampling.

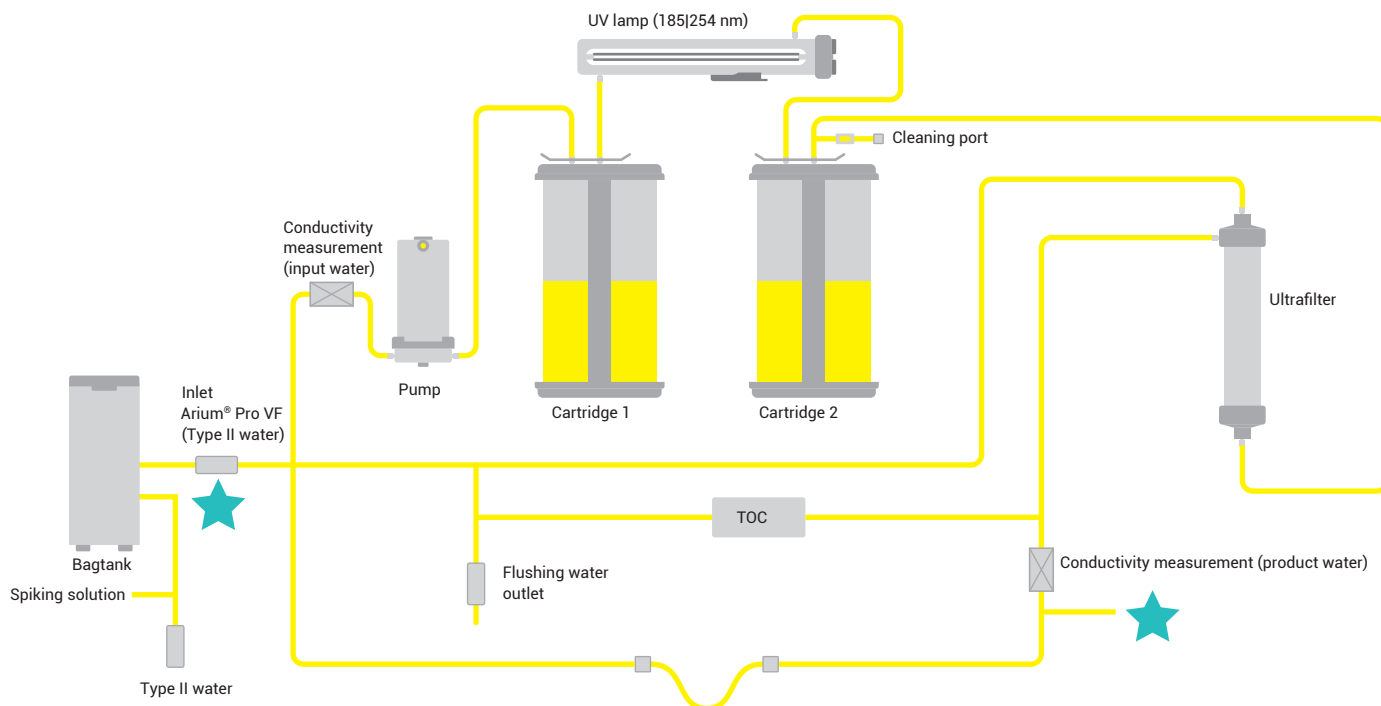


Figure 3: Flowchart of Arium® Pro VF for the first objective. Teal color stars mark the point of sampling.

RESULTS

The workload was divided into two separate experimental setups. The first consisted of: one tap-water sample, one type II water sample and one Arium®-purified type I water sample (in total three 1L samples, Figure 3). The second setup (Figure 2) consisted of: one type II water sample spiked with 10, 25 and 70 µm polystyrene (PS) microspheres, one Arium®-purified water sample obtained after consuming 9, 14 and 19 L of PS microsphere-spiked type II water (three separate samples) and finally two Arium purified water samples, where the first 1 L sample was taken after consuming 3 L of non-spiked type II water, and the other after 5 L (in total six 1 L samples). Type II water used in the first setup was poured directly in a clean 1 L laboratory glass, while the type II water used in the second setup was fed to the Arium device using a plastic canister. All experiments were performed twice, bringing the total number of experiments to 18.

Next step was to determine method limit of detection (LOD), in terms of numbers of microplastics per type of polymer and per particle size range. This was done by calculating the sum of mean number of particles from seven analyses of 1 L procedural blank samples and three standard deviations.

All water samples were processed as follows: Samples were vacuum filtered through silicon filters with 1 µm pore size, which were subsequently stored in an appropriate container, to avoid particle contamination. Individual filters were placed onto a filter holder, which was then mounted onto a microscope stage and brought to objective focus.

Image of the entire filter surface was obtained by the instrument-controlled camera, after which the individual particles on the filter surface were located using the instrument software. Finally, the particles were analyzed using Raman micro-spectroscopy, yielding information regarding the number, size, and chemical identity of individual particles. Spectral match of a particle spectrum with a spectrum in a spectral library of minimum 70 % was taken as a criterium for including a particle in the reported results.

Limit of detection (LOD) needed to be calculated in order to be able to add context to later results obtained in two experimental setups. In the case of microplastics analyses, method LOD represents the smallest number of particles of a certain polymer type and size range that can be detected using the said analytical method. In this work, seven procedural blank samples were prepared and analyzed, after which the results were used for calculating the LOD:

$$\text{LOD} = \text{mean}_{7 \text{ samples}} + 3 \times \sigma_{7 \text{ samples}}$$

Results of seven analyses, along with the calculated LOD, are shown in Table 1. Particles detected in the blank samples consisted of seven types of polymers: polyethylene (PE), polypropylene (PP), polytetrafluoroethylene (PTFE), polystyrene (PS), ethylene vinyl acetate (EVA), polyamide (PA) and polyethylene terephthalate (PET) (Figure 4).

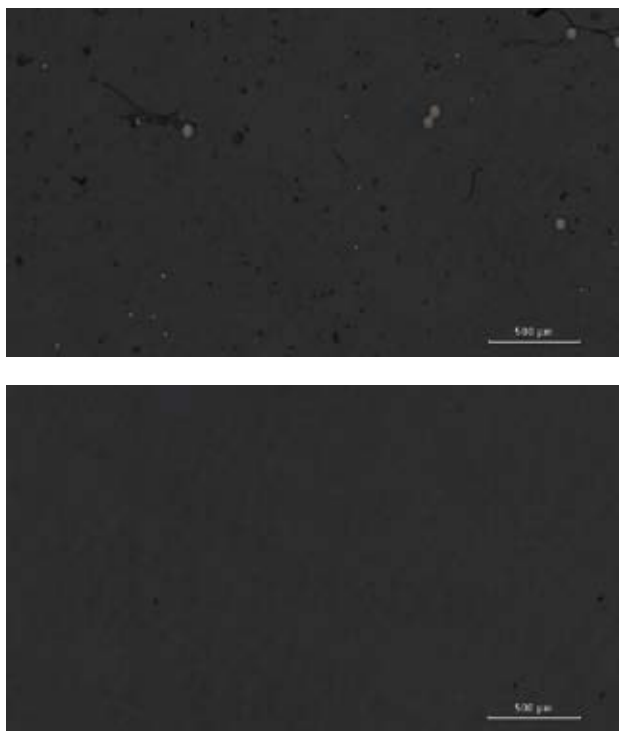


Figure 4: Filter with microplastic particles (top) and filter without any microplastic particles (bottom).

Polymer Type	Size Range (µm)	Number of Particles									
		1	2	3	4	5	6	7	Average	3 x σ7	LOD
PE	1-50	1	0	1	1	0	4	1	1.1	4.0	5.2
	50-100	1	1	2	0	0	1	1	0.9	2.1	2.9
	100-500	0	0	0	0	0	0	0	0.0	0.0	0.0
	> 500	0	0	0	0	0	0	0	0.0	0.0	0.0
PP	1-50	0	0	0	1	0	0	0	0.1	1.1	1.3
	50-100	1	0	0	0	1	0	2	0.6	2.4	2.9
	100-500	2	1	1	1	3	1	5	2.0	4.6	6.6
	> 500	0	0	0	0	1	0	0	0.1	1.1	1.3
PS	1-50	1	0	0	0	0	0	0	0.1	1.1	1.3
	50-100	0	0	0	0	0	0	0	0.0	0.0	0.0
	100-500	0	0	0	0	0	0	0	0.0	0.0	0.0
	> 500	0	0	0	0	0	0	0	0.0	0.0	0.0
PET	1-50	0	0	0	0	0	0	0	0.0	0.0	0.0
	50-100	0	1	0	0	0	0	0	0.1	1.1	1.3
	100-500	0	0	0	0	0	0	0	0.0	0.0	0.0
	> 500	0	0	0	0	0	0	0	0.0	0.0	0.0
EVA	1-50	0	0	0	0	0	1	0	0.1	1.1	1.3
	50-100	0	0	0	0	0	0	0	0.0	0.0	0.0
	100-500	0	0	0	0	0	0	0	0.0	0.0	0.0
	> 500	0	1	0	0	0	0	0	0.1	1.1	1.3
PA	1-50	0	0	0	0	0	1	1	0.3	1.5	1.7
	50-100	0	0	0	0	0	0	0	0.0	0.0	0.0
	100-500	0	0	0	0	0	0	0	0.0	0.0	0.0
	> 500	0	0	0	0	0	0	0	0.0	0.0	0.0
PTFE	1-50	0	0	0	0	0	0	0	0.0	0.0	0.0
	50-100	0	0	0	0	0	1	0	0.1	1.1	1.3
	100-500	0	0	0	0	0	3	1	0.6	3.4	4.0
	> 500	0	0	0	0	0	0	0	0.0	0.0	0.0
Cellulose	1-50	0	0	0	1	2	3	5	1.6	5.7	7.3
	50-100	0	0	0	0	0	6	0	0.9	6.8	7.7
	100-500	3	2	0	0	0	2	2	1.3	3.8	5.0
	> 500	0	0	0	0	0	3	1	0.6	3.4	4.0

Table 1: Results for the analyses of seven procedural blank samples, along with the calculated LOD.

First set of experiments, which consisted of one tap-water sample, one type II water sample and one Arium®-purified type I water sample (see Figure 2), was done in order to see if the Arium® purification device leached microplastics into the product water.

Results of the analyses are shown in the Table 2. These results show that all three types of water samples, contained a number of particles smaller than the method LOD. This provides the necessary evidence to state that the Arium system does not leach a detectable amount of microplastic particles into the water it produces.

Polymer Type	Size Range (µm)	1st Experimental Setup								
		Tap Water			Type II Water			Arium® Water		
		Number of Particles			Number of Particles			Number of Particles		
		1	2	Average	1	2	Average	1	2	Average
PE	1- 50	0	6	3	2	0	1	4	1	2.5
	50 - 100	0	1	0.5	0	0	0	1	1	1
	100 - 500	1	0	0.5	0	0	0	0	0	0
	> 500	0	0	0	0	0	0	0	0	0
PP	1- 50	0	1	0.5	0	0	0	0	0	0
	50 - 100	0	1	0.5	2	1	1.5	0	2	1
	100 - 500	0	1	0.5	2	5	3.5	1	5	3
	> 500	0	0	0	0	0	0	0	0	0
PTFE	1- 50	0	1	0.5	0	0	0	0	0	0
	50 - 100	0	0	0	0	0	0	1	0	0.5
	100 - 500	0	0	0	0	0	0	3	1	2
	> 500	0	0	0	0	0	0	0	0	0
PS	1- 50	0	0	0	2	0	1	0	0	0
	50 - 100	0	0	0	0	0	0	0	0	0
	100 - 500	0	0	0	0	0	0	0	0	0
	> 500	0	0	0	0	0	0	0	0	0
EVA	1- 50	1	0	0.5	0	0	0	1	0	0.5
	50 - 100	0	0	0	0	0	0	0	0	0
	100 - 500	1	0	0.5	0	0	0	0	0	0
	> 500	0	0	0	0	0	0	0	0	0
PA	1- 50	0	0	0	0	0	0	1	1	1
	50 - 100	0	0	0	0	0	0	0	0	0
	100 - 500	0	0	0	0	0	0	0	0	0
	> 500	0	0	0	0	0	0	0	0	0
Cellulose	1- 50	0	0	0	0	1	0.5	3	5	4
	50 - 100	0	1	0.5	0	1	0.5	6	0	3
	100 - 500	1	0	0.5	1	3	2	2	2	2
	> 500	0	0	0	0	0	0	3	1	2

Table 2: Results for the first experimental setup (refer Figure 2).

Second set of experiments was performed in order to see if the system retained the particles introduced to the system from the feed water (refer Figure 3). The setup consisted of one type II water sample spiked with 10, 25 and 70 μm polystyrene (PS) microspheres, one Arium[®]-purified water sample obtained after consuming 9, 14 and 19 L of PS microsphere-spiked type II water (three separate samples) and finally two Arium[®]-purified water samples, where the first 1 L sample was taken after consuming 3 L of non-spiked type II water, and the other after 5 L. Results of the analyses of samples in the second experimental setup are shown in Table 3.

Results for the two analyses of water spiked with PS particles show that PS particles were clearly detected before subsequent water purification. In contrast to these results, analyses of Arium[®]-purified water samples obtained after consuming 9, 14 and 19 L of PS microsphere-spiked type II water showed that PS particles do not get transferred into the final product water. Furthermore, results show that even after swapping PS-spiked type II water with clean type II water, the PS microparticles are not being washed out of the system. This indicates that the Arium[®] purification system retains all particles introduced to the system and prevents them from ending up in the final product.

Polymer Type	Size Range (μm)	2nd Experimental Setup											
		Type II water + PS			Purified spiked water after 9L			Purified spiked water after 14L			Purified spiked water after 19L		
		Number of Particles			Number of Particles			Number of Particles			Number of Particles		
		1	2	Average	1	2	Average	1	2	Average	1	2	Average
PS	1- 50	115	53	84	0	0	0	0	0	0	0	0	0
	50 - 100	44	49	46.5	0	0	0	0	0	0	0	0	0
	100 - 500	20	0	10	0	0	0	0	0	0	0	0	0
	> 500	0	0	0	0	0	0	0	0	0	0	0	0

Table 3: Results for the second experimental setup – spiking experiments (refer Figure 3).

Polymer Type	Size Range (μm)	2nd Experimental Setup					
		Purified type II water after 3L			Purified type II water after 5L		
		Number of Particles			Number of Particles		
		1	2	Average	1	2	Average
PS	1- 50	0	0	0	0	0	0
	50 - 100	0	0	0	0	0	0
	100 - 500	0	0	0	0	0	0
	> 500	0	0	0	0	0	0

Table 4: Results for the second experimental setup – washing the system with clean type II water.

RETURN ON INVESTMENT

The cost analysis provided over a one-year period, as shown in Figure 5 clearly highlights the advantages of transitioning to in-house treated water. With a daily demand of two litres, the investment in the treatment device starts to yield significant financial benefits in as little as five months, ultimately resulting in substantial cost savings within the course of a year. Furthermore, this approach aligns with sustainability goals by eliminating the need for glass bottles, reducing pollution associated with shipping, and eliminating the necessity for extensive storage space.

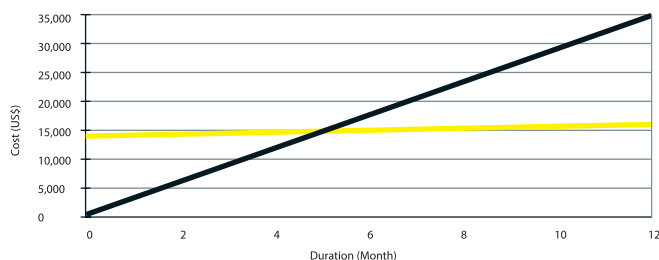


Figure 5: Comparison of expenses between In-house produced Arium®-water Type I (ultrapure water, yellow) and bottled water (VWR: Water Reag. Ph. Eur. 1095507 particle-free, 21.11.2023; 72 US\$ per liter, black). Calculation based on following assumptions: two liters water consumption per working day, 20 working days per month, Arium® Pro VF system plus consumables and annual consumables (list price) and costs of bottled water.

CONCLUSION

Two different kinds of testing protocols were tested with the Arium® Pro VF system. The first test setup showed that no microplastic contaminations were leached out from the device when deionized water was introduced to the machinery.

The Arium® Pro VF system also removed microplastic particles efficiently from water which was purposely spiked with high amount of microplastics.

The Arium® Pro VF system can be used when laboratories need to be extra sure that no unwanted microplastic particles can contaminate the water used in the laboratory.

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